DYNAMICAL SYSTEMS: FROM THEORY TO APPLICATIONS

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Report Approval

This dissertation entitled **Dynamical Systems: From Theory To Applications** by **Arpan Saha** (**Examination Roll No.: 6R24MTMA2005, Registration No.: A04-1112-0234-21**) of undergraduate sixth semester is approved for the partial fulfilment of the degree of **Bachelors of Science Honours in Mathematics** from **Ramakrishna Mission Vidyamandira**.

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Declaration

I affirm that this project work is original with citations to the appropriate references, and has not been submitted for any other degree or qualification. I adhere to the ethical standards of academic integrity and have properly cited all sources used. By signing this declaration, I attest to the authenticity and integrity of the work presented in this dissertation.

Arpan Saha

Abstract

Dynamical system is a strong framework which we can use to deal with simple real world problems to complex cosmological problems. This dissertation starts with a brief history of dynamics, then with a strong focus on key topics like non-linear differential equations, fixed Points and stability, phase portraits, linear stability theory, flows on line and plane, and the bifurcations in various systems. Then it focuses on various cosmological equations describing the dynamics of the universe which are mainly systems of ordinary differential equations, and one of the most elegant ways these can be investigated is by casting them into the form of dynamical systems. This dissertation emphasizes on the equations governing the behaviour of a cosmological model, resulting in the desire to analyze them, and then use some beautiful transformations to make them recognisable to us in the form of autonomous systems. Then applying our theories, we will understand the behaviours, sometimes by pictures also. In conclusion, this dissertation uses dynamical systems as a powerful tool to study the cosmological nature of the universe and thereby create new prospects for future research in this subject.

Acknowledgements

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Sincerely,
Arpan Saha

Contents

Chapter 1

Basics of Dynamical Systems

1.1 Introduction

Dynamical systems theory is a very strong framework which deals with the changes in various systems over time subject to some beautiful conditions. Robert Luke Devaney (American mathematician) once said, " Modern dynamical systems theory has a relatively short history. It begins with Poincare (of course) ... [to whom] a global understanding of the gross behaviour of all solutions of the system was more important than the local behaviour of particular, analytically precise solutions." Through the pillar components - critical points, phase portraits, eigenvalue analysis etc. it gives us beautiful patterns of the behaviours of complex systems. The aristocracy of dynamical systems lies not only in its theoretical beauty but also in its practical applications. Whether in predicting weather patterns, optimising resource allocation in ecosystems or exploring the behaviour of the universe, dynamical systems theory offers valuable pieces of information to us regarding these fields. Throughout this dissertation work, the fundamental components of dynamical systems have been discussed along with their insightful applications to various cosmological models, thereby analyzing some beautiful patterns in those systems through the help of phase portraits.

1.2 A Short History of Dynamical Systems

The modern theory of dynamical systems derives from the work of H. J. Poincaré (1854-1912) on the three-body problem of celestial mechanics and primarily from a single, massive and initially flawed paper. He introduced a new point of view that emphasized qualitative rather than quantitative questions. To explain it properly, we can say that instead of asking for the exact positions of the planets at all times, he asked, "Is the solar system stable forever, or will some planets eventually fly off to infinity?" Answering this question Poincaré devised a new way of analysis which emphasized the qualitative approach. This eventually gave birth to the subject 'Dynamical Systems'. Following Poincaré's work, J. Hadamard (1865-1963) considered the dynamics of geodesic flows, but the next major thrust was due to G. D. Birkhoff (1884-1944), who early in his career had proved Poincaré's "last theorem" on fixed points of annulus maps. The other pioneers in this area include Van der Pol, Andronov, Littlewood, Cartwright, Levinson and Smale. Later, although chaos stole the spotlight, Winfree applied the geometric methods of dynamics to biological oscillations, especially circadian (roughly 24-hour) rhythms and heart rhythms which made a breakthrough in the emerging area of mathematical biology. By the 1980s many people were working on dynamics, with contributions too numerous to list.

1.3 Theoretical Fundamentals of Dynamical Systems

This section focuses on theoretical foundations and key terms of dynamical systems. The core fundamental components of dynamical systems are described below:

Definition 1. (Dynamical Systems). *Dynamical system is an abstract system consisting of*
1. a space (state space or phase space), and
2. a mathematical rule describing the evolution of points in that space.

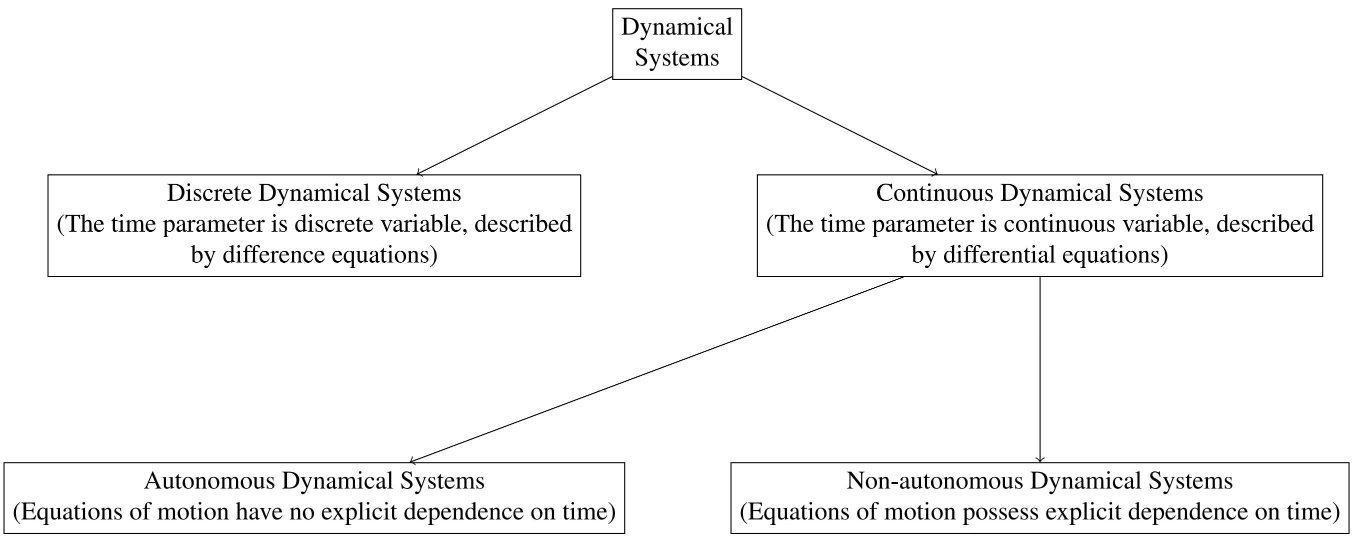

Figure 1.1: Classification of Dynamical Systems

In this dissertation, we are going to work with autonomous dynamical systems. So, let us define it mathematically and develop the whole business.

Definition 2. (Autonomous Dynamical Systems). *Let* $\mathbf{x} = (x_1(t), x_2(t), ..., x_n(t)) \in X \subseteq \mathbb{R}^n$. *An autonomous dynamical system is written in the form,*

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \tag{1.1}$$

where, $\mathbf{f} : X \to X$ *is sufficiently smooth, dot denotes differentiation with respect to some suitable time parameter* $t \in \mathbb{R}$. *The function* $\mathbf{f}$ *can be viewed as a vector field on* $\mathbb{R}^n$ *such that* $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), ..., f_n(\mathbf{x}))$.

Example 1. A simple example of autonomous system is,

$$\frac{dx}{dt} = 3x + 4y, \tag{1.2a}$$

$$\frac{dy}{dt} = 7x + 9y, \tag{1.2b}$$

where, $t \in \mathbb{R}$.

Definition 3. (Fixed Points). *The autonomous system* $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ *is said to have a fixed (or stable or equilibrium) point at* $\mathbf{x} = \mathbf{x_0}$ *if* $\mathbf{f}(\mathbf{x_0}) = \mathbf{0}$.

Definition 4. (Stable & Unstable Fixed Points). *Let* $\mathbf{x_0}$ *be a fixed point of the system (1.1). It is called stable if for every* $\epsilon > 0$*, we can find a* $\delta > 0$ *such that if* $\psi(t)$ *is any solution of (1.1) satisfying* $||\psi(t_0) - \mathbf{x_0}|| < \delta$*, then the solution* $\psi(t)$ *exists for all* $t \geq t_0$ *and it will satisfy* $||\psi(t) - \mathbf{x_0}|| < \epsilon$ *for all* $t \geq t_0$.

Definition 5. (Asymptotically Stable Fixed Points). *Let* $\mathbf{x_0}$ *be a stable fixed point of the system (1.1). It is called asymptotically stable if there exists a number* $\delta > 0$ *such that if* $\psi(t)$ *is any solution of (1.1) satisfying* $||\psi(t_0) - \mathbf{x_0}|| < \delta$*, then* $lim_{t\to\infty}\psi(t) = \mathbf{x_0}$.

The subtle difference between **Definition 4** & **Definition 5** is that all trajectories near an asymptotically stable fixed point will eventually reach that point, while trajectories near a stable fixed point could, for instance, circle around that point.

To observe a dynamical system over time we need a collection of data taken from an authentic source. Also, we need to analyze them pictorially to have a very clear idea regarding them. For this, we need some special terms called phase space, phase portrait and time series analysis. Phase space is a multidimensional space which contains the complete behaviour of each state of a dynamical system. Each point in the space corresponds to a unique state of a dynamical system. Here we will work with the two-dimensional systems for which the phase space turns into a plane called the phase plane. The trajectories of the systems corresponding to the space are called phase portraits. The evolution of the systems over time is determined by plotting the different states of a system with respect to different times. This is known as the time series of the dynamical systems.

Definition 6. (Basin of Attraction). *It is the collection of initial conditions in phase space such that the trajectories started from them approach the sink in the limit* $t \to \infty$*. In the figure described below, the regions to the left and right of the curve* AF_1B *are the basins of attraction of the equilibrium points* F_2 *and* F_3 *respectively. The curve* AF_1B *is the (basin) boundary separating the basins of attraction of the two stable equilibrium points.*

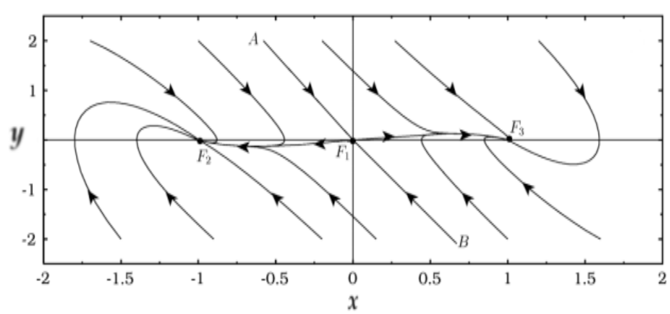

Figure 1.2: Basins of Attraction

Definition 7. (Stability Theory). *Stability theory addresses the stability of solutions of differential equations and trajectories of dynamical systems under small perturbations of initial conditions.*

Definition 8. (Bifurcations). *The characteristics of a dynamical system depend upon several parameters. For a small change in those parameters, considerable changes can happen in a system. The parameters are called critical parameters and this phenomenon is known as Bifurcation.*

Transformation of non-autonomous systems to autonomous systems: For non-autonomous systems, sometimes we can formally introduce a new dependent variable and rewrite the system as a set of autonomous systems. e.g. consider the non-autonomous dynamical system,

$$\frac{dx}{dt} = y, \tag{1.3a}$$
$$\frac{dy}{dt} = -\alpha y - \omega_0^2 x - \beta x^3 + f \cos \omega t. \tag{1.3b}$$

We introduce a new dependent variable $z = \omega t$ in the above system. Then the equivalent first-order autonomous dynamical system is:

$$\frac{dx}{dt} = y, \tag{1.4a}$$
$$\frac{dy}{dt} = \alpha y - \omega_0^2 x - \beta x^3 + f \cos z, \tag{1.4b}$$
$$\frac{dz}{dt} = \omega. \tag{1.4c}$$

1.4 One-Dimensional Flows

To explore the complex dynamical systems, firstly, we have to start with the basics. The advantage of analyzing one-dimensional systems is that these can be verified with their actual solutions. In this chapter, we will discuss $flows\ on\ the\ line$ and $flows\ on\ the\ circle$.

1.4.1 Flows on the Line

Introduction

Previously, we have defined the dynamical systems in a general way. Here we will start with the case when $n = 1$. Then we get a single equation of the form

$$\dot{x} = f(x). \tag{1.5}$$

Here, $x(t)$ is a real-valued function of time t, and $f(x)$ is a smooth real-valued function of x. We will call such equations ***one-dimensional*** or ***first-order systems.*** Before there is any chance of confusion, we discuss two fussy points of terminology:

1. The word $system$ is being used here in the sense of a dynamical system, not in the classical sense of a collection of two or more equations. Thus a single equation can be a $system$.

2. We allow only $autonomous$ systems in here. The $non\text{-}automonous$ systems are much more complicated and can be treated differently.

A Geometric Way of Thinking

Along with our discussion, we will introduce one of the most basic techniques of dynamics: *interpreting a differential equation as a vector field.* Consider the following nonlinear differential equation:

$$\dot{x} = \sin x, \tag{1.6}$$

Now, given an initial condition $x = x_0$ at $t = 0$, the solution is,

$$t = \ln \left| \frac{\csc x_0 + \cot x_0}{\csc x + \cot x} \right|. \tag{1.7}$$

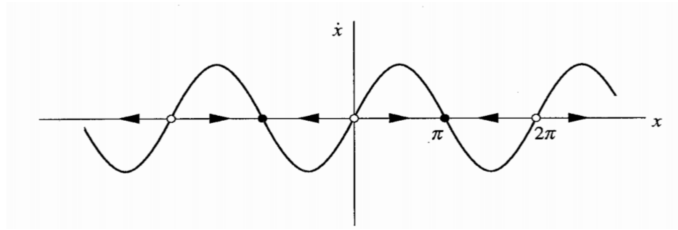

Figure 1.3: Graphical Analysis

Now, we see that this formula is not transparent. It is unable to answer many questions. In contrast, we see the graphical analysis of (1.7) is clear and simple.
As shown in Figure 1.3, the $flow$ is to the right when $\dot{x} > 0$ and to the left when $\dot{x} < 0$. At points where $\dot{x} = 0$, there is no flow. These points are called *fixed points*. The solid black dots are $stable$ fixed points and open circles are $unstable$ fixed points. Armed with all these concepts, we can sketch the qualitative form of the solution as follows:

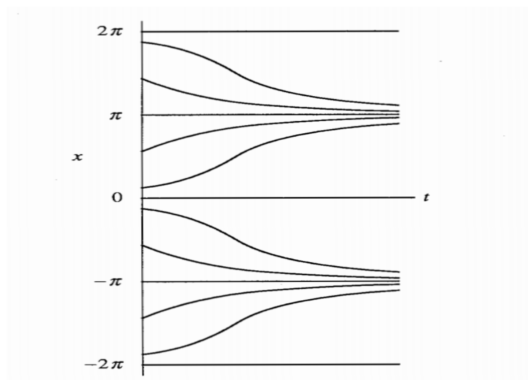

Figure 1.4: Qualitative Behaviour

Linear Stability Analysis

To have a more quantitative measure of stability, such as the rate of decay to a stable fixed point, we have to explore other methods apart from the graphical analysis. One of the best among those is $linearizing$ about a fixed point, as we now explain.
let x^* be a fixed point. Consider a small perturbation in the equation (1.5),

$$\eta(t) = x(t) - x^*, \tag{1.8}$$

As x^* is constant, differentiating both sides w.r.t. t,

$$\dot{\eta} = \dot{x} = f(x) = f(x^* + \eta), \tag{1.9}$$

Now using Taylor series expansion, we obtain

$$f(x^* + \eta) = f(x^*) + \eta f'(x^*) + O(\eta^2), \tag{1.10}$$

where $O(\eta^2)$ denotes quadratically small terms in η. Now, $f(x^*) = 0$, as x^* is a fixed point. Hence,

$$\dot{\eta} = \eta f'(x^*) + O(\eta^2), \tag{1.11}$$

if $f'(x^*) \neq 0$, the $O(\eta^2)$ terms are negligible and we get,

$$\dot{\eta} \approx \eta f'(x^*), \tag{1.12}$$

The equation (1.12) is called $linearization$ of the system (1.5) about x^*. The system becomes unstable if $f'(x^*) > 0$, and becomes stable if $f'(x^*) < 0$. If $f'(x^*) = 0$, the $O(\eta^2)$ terms are not negligible and a nonlinear analysis is needed to determine stability.

Existence and Uniqueness

To study the systems of the form (1.1), we must have to check the existence and uniqueness of solutions in the desired domain, otherwise our whole analysis will collapse. We state an important theorem regarding this issue in the following:

Theorem 1. ***(Existence and Uniqueness Theorem)****. Consider the initial value problem of the form (1.5) with initial condition $x(0) = x_0$. Suppose that $f(x)$ and $f'(x)$ are continuous on an open interval R of the x-axis, and suppose that x_0 is a point in R. Then the initial value problem has a solution $x(t)$ on some interval $(-\tau, \tau)$ about $t = 0$, and the solution is unique.*

Dimensional Analysis and Scaling

In the systems of 2nd or higher order, or when there are too many parameters, it is helpful to express the equation into $dimensionless$ form as nondimensionalizing the equation reduces the number of parameters by lumping them together into dimensionless groups.

A Paradox

In some cases, after executing dimensional analysis, a 2nd order differential equation is transformed into a first-order equation (under some conditions). But this is something fundamentally wrong as in our first case, we have two initial conditions. How can we guarantee that the first-order equation will satisfy the two initial conditions?

Phase Plane Analysis

To solve the paradox, we do a phase plane analysis where we break down the 2nd-order equation into two first-order equations and do the analysis (detailed discussion in next chapters). e.g. Consider the equation,

$$\epsilon \frac{d^2\phi}{d\tau^2} = -\frac{d\phi}{d\tau} - \sin\phi + \gamma\ \sin\phi\ \cos\phi, \tag{1.13}$$

Introducing a new variable Ω, we write equation (1.13) as,

$$\phi' = \Omega, \tag{1.14a}$$

$$\Omega' = \frac{1}{\epsilon}(f(\phi) - \Omega), \tag{1.14b}$$

Thus, we break down it into a system of first-order equations and now be able to do the phase plane analysis.

1.4.2 Flows on the Circle

This time we consider a new kind of differential equation,

$$\dot{\theta} = f(\theta), \tag{1.15}$$

which corresponds to a ***vector field on the circle.*** Here, θ is a point on the circle and $\dot{\theta}$ is the velocity vector at that point, determined by the equation (1.15). Like the line, the circle is one-dimensional, but it has an important new property: by flowing in one direction, a particle can eventually return to its starting place. To put it another way, *vector fields on the circle provide the most basic model of systems that can oscillate.* However, in all other prospects, flows on the circle are similar to flows on the line.

1.5 Two-Dimensional Flows

1.5.1 Linear Systems

As we have seen, in one-dimensional phase spaces the flow is extremely confined, all trajectories are forced to move monotonically or remain constant. But in higher dimensions, trajectories have much more options to move. So we have to upgrade our machinery. First, we will start with linear systems and then gradually uplift our idea to nonlinear systems.

Definition 9. (Linear System) A *two-dimensional linear system* is a system of the form

$$\dot{x} = ax + by, \tag{1.16a}$$
$$\dot{y} = cx + dy, \tag{1.16b}$$

where a, b, c, d are parameters. This system can also be written more compactly in matrix form as

$$\dot{X} = AX, \tag{1.17}$$

where A = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and X = $\begin{pmatrix} x \\ y \end{pmatrix}$.

Example 2. The vibrations of a mass hanging from a linear spring are governed by the system

$$\dot{x} = \nu, \tag{1.18a}$$
$$\dot{\nu} = -\frac{k}{m}x, \tag{1.18b}$$

where m is the mass, k is the spring constant and x is the displacement of the mass from equilibrium.

Now for nonlinear systems, there is typically no hope of finding the trajectories analytically. Instead, we will try to determine the $qualitative$ behaviour of the solutions.

1.5.2 Nonlinear Systems

Generally, nonlinear systems are those which are not linear. In our study we will work with systems of the type (1.1) with $n = 2$.

Existence and Uniqueness

We have been a bit optimistic so far. At this stage, we have no guarantee that the system of type (1.1) even has solutions! So we generalize **Theorem 1** to two-dimensional systems. We state the result for n-dimensional systems since no extra effort is involved:

Theorem 2. ***(Existence and Uniqueness Theorem for higher dimensions)****.*
Consider the initial value problem

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \tag{1.19}$$

with the initial condition $\mathbf{x}(0) = \mathbf{x}_0$*. Suppose that* $\mathbf{f}$ *is continuous and that all its partial derivatives* $\frac{\partial f_i}{\partial x_j}$*, where* i, j = $1, ..., n$ *are continuous for* $\mathbf{x}$ *in some open connected set* $D \subset \mathbb{R}^n$*. Then for* $\mathbf{x_0} \in D$*, the initial value problem has a solution* $\mathbf{x}(t)$ *on some time interval* $(-\tau, \tau)$ *about* $t = 0$*, and the solution is unique.*

1.5.3 Fixed Points and Linearization

Consider a two-dimensional dynamical system,

$$\frac{dx}{dt} = P(x, y), \tag{1.20a}$$
$$\frac{dy}{dt} = Q(x, y), \tag{1.20b}$$

Where P and Q have continuous first partial derivatives for all x and y and they does not depend explicitly on the variable t. Let $X = X^* = X_0 \equiv (x_0, y_0)$ be the equilibrium point of the above system. Then at this point, the system does not change its state. So, $\dot{x} = \dot{y} = 0$, i.e. $P(x_0, y_0) = Q(x_0, y_0) = 0$. In order to determine stability, we slightly disturb it (or we say that we infinitesimally perturb it),

$$x = x_0 + \xi(t), \tag{1.21a}$$
$$y = y_0 + \eta(t), \tag{1.21b}$$

where $\xi, \eta \ll 1$. Now to determine the resultant motion, we use Taylor series expansion of the functions about the equilibrium point,

$$P(x, y) = P(x_0 + \xi, y_0 + \eta) = P(x_0, y_0) + \tfrac{\partial P}{\partial x}|_{(x_0,y_0)} \cdot \xi + \tfrac{\partial P}{\partial y}|_{(x_0,y_0)} \cdot \eta + \text{higher order terms in } (\xi, \eta),$$
$$Q(x, y) = Q(x_0 + \xi, y_0 + \eta) = Q(x_0, y_0) + \tfrac{\partial Q}{\partial x}|_{(x_0,y_0)} \cdot \xi + \tfrac{\partial Q}{\partial y}|_{(x_0,y_0)} \cdot \eta + \text{higher order terms in } (\xi, \eta).$$

Now using the fact that at the equilibrium point $P(x_0, y_0) = Q(x_0, y_0) = 0$, and neglecting the higher order terms (since they are tiny), the two-dimensional dynamical system can be rewritten as,

$$\dot{\xi} = \dot{x} = a\xi + b\eta, \tag{1.22a}$$
$$\dot{\eta} = \dot{y} = c\xi + d\eta, \tag{1.22b}$$

where, a, b, c, d are constants and given by,

$$a = \tfrac{\partial P}{\partial x}|_{(x_0,y_0)},\ b = \tfrac{\partial P}{\partial y}|_{(x_0,y_0)},$$
$$c = \tfrac{\partial Q}{\partial x}|_{(x_0,y_0)},\ d = \tfrac{\partial Q}{\partial y}|_{(x_0,y_0)}.$$

Therefore the system evolves to the matrix of the form,

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix}, \tag{1.23}$$

$$\dot{X} = AX, \tag{1.24}$$

where $\dot{X} = \begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix}$, $X = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the ***Jacobian*** matrix of the ***linearized*** system.
Using standard methods the solution of the ***linearized*** system is,

$$\xi(t) = A\, e^{(\lambda_1 t)} + B\, e^{(\lambda_2 t)},$$
$$\eta(t) = C\, e^{(\lambda_1 t)} + D\, e^{(\lambda_2 t)},$$

where A, B, C, D are constants and λ_1, λ_2 are eigen values of the matrix A.
Thus the stability nature of an equilibrium points is solely dependent on the eigen values of the matrix. Relation between the stability nature of the equilibrium point and the eigenvalues are described in the following:
(a) **Stable:** Real parts of both the eigenvalues are negative.
(b) **Unstable:** Real part of even one of the two eigenvalues is positive.
(c) **Neutral:** Real parts of both the eigenvalues are zero.

Effect of The Small Nonlinear Terms

Is it really safe to omit the nonlinear terms of higher order in the process of linearization? Is the qualitative nature of the linearized system the same as that of the original system? The answer is yes. As long as the system is not like the cases when det(A) = 0, the fixed equilibrium point is the same as predicted by the linearized system (See Andronov et al. (1973) for proof of this result). The borderline cases are much more delicate. They can be altered by small nonlinear terms.

1.5.4 Classification of Equilibrium Points

Case 1: $\lambda_1 \leq \lambda_2 < 0$ (Stable Node/ Star)

In this case the equilibrium point is stable. Thus any trajectory starting from the neighbourhood of the equilibrium point (x_0, y_0) in the $(x$ - $y)$ phase plane approaches it exponentially in the long time limit. To study how the trajectories approach the equilibrium point, we consider the slope dy/dx of the trajectories. Shifting the origin to the equilibrium point and rewriting the equations for new variables we get,

$$\frac{dy}{dx} = \frac{C\lambda_1 e^{(\lambda_1-\lambda_2)t} + D\lambda_2}{A\lambda_1 e^{(\lambda_1-\lambda_2)t} + B\lambda_2}.$$

Now, we have to consider two cases.
(i) $\lambda_1 = \lambda_2$:
In this case, $\frac{dy}{dx} = \frac{C\lambda_1 + D\lambda_2}{A\lambda_1 + B\lambda_2} = m$ (say). So it will become a straight line which passes through the equilibrium point. This type of equilibrium point is called *stable star.*

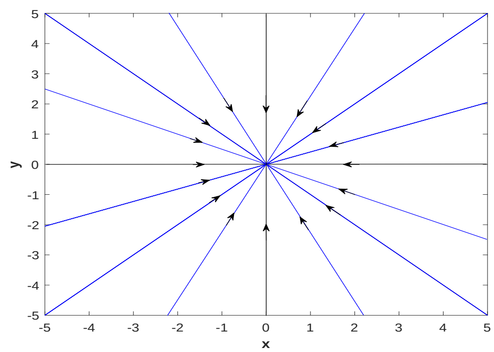

Figure 1.5: Stable Star

(ii) $\lambda_1 < \lambda_2$: Then the trajectories approach the equilibrium point in a parabolic path. This type of equilibrium point is called *stable node.*

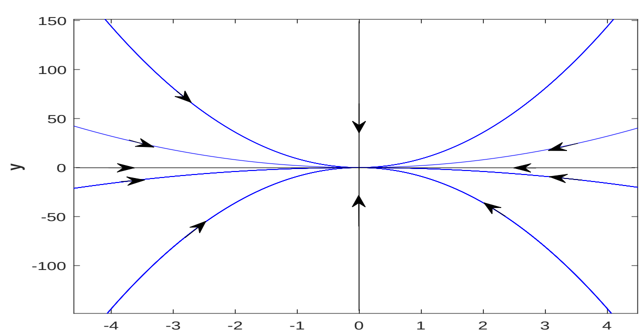

Figure 1.6: Stable Node

Case 2: $\lambda_1 \geq \lambda_2 > 0$ **(Unstable Node/Star)**

In this case trajectories diverge from the equilibrium point. For $\lambda_1 = \lambda_2$, the equilibrium point will be *unstable star*. For $\lambda_1 > \lambda_2$, the equilibrium point will be *unstable node.*

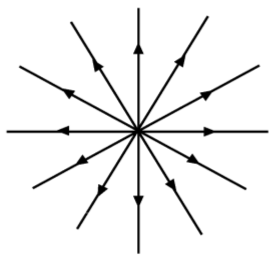

Figure 1.8: Unstable Star

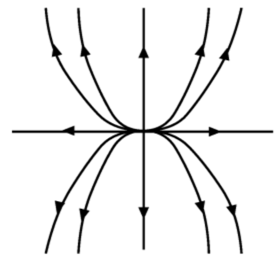

Figure 1.9: Unstable Node

Case 3: λ_1, λ_2 **complex conjugates (Stable/Unstable Focus)**

Let λ_1 and λ_2 are complex conjugates given by
$\lambda_1 = \alpha + i\beta$, $\lambda_2 = \alpha - i\beta$ where α, β are real constants and $\beta \neq 0$. Therefore the solution will be

$$\xi(t) = |A|\, e^{\alpha t} \cos(\beta t + \phi),$$
$$\eta(t) = |D|\, e^{\alpha t} \sin(\beta t + \phi_1),$$

where, A, D are constants and ϕ, ϕ_1 are angles given by

$$A = A_r + iA_i,\, D = D_r + iD_i,$$
$$\phi = \tan^{-1}(A_i/A_r),\, \phi_1 = \tan^{-1}(D_r/D_i).$$

Now, we consider two cases.
(i) $\alpha < 0$:
Both $|\xi|$ and $|\eta| \to 0$ as $t \to \infty$. Then the equilibrium point is stable. All the trajectories move to the point spirally before meeting with it asymptotically. Then the equilibrium point is called *stable focus*.
(ii) $\alpha > 0$:
The trajectories near to it moves away from it as $t \to \infty$. In that case the equilibrium point is called *unstable focus.*

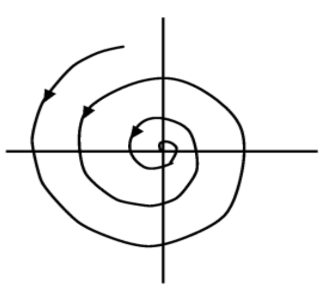

Figure 1.10: Stable Focus

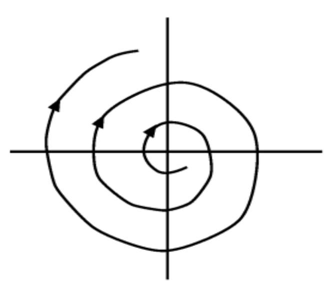

Figure 1.11: Unstable Focus

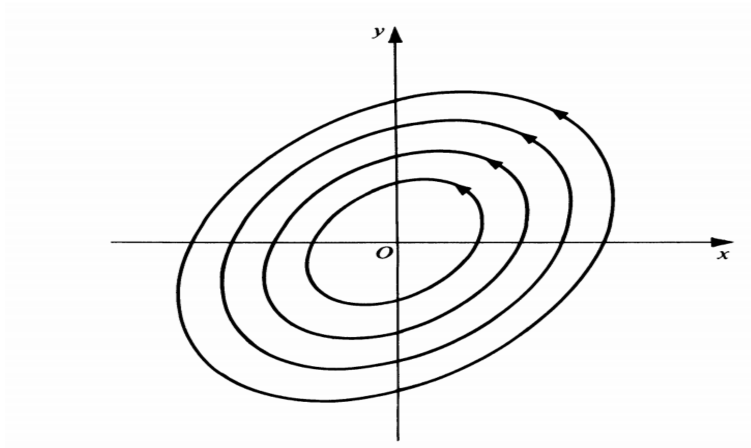

Figure 1.12: Center

Case 4: λ_1 and λ_2 pure imaginary and complex conjugates (Center/Elliptic)

This is the case when $\alpha = 0$. Then $\lambda_{1,2} = \pm i\beta$. So,

$$\xi(t) = |A| \cos{(\beta t + \phi)},$$
$$\eta(t) = |D| \sin{(\beta t + \phi_1)}.$$

Then the trajectories neither move towards it nor move away from it as $t \to \pm\infty$. This equilibrium point is called *center type* or *elliptic equilibrium point* and is neutrally stable. (To study its true stability properties, we have to consider the neglected nonlinear terms.)

Case 5: $\lambda_1 < 0 < \lambda_2$ (Saddle Point)

For any one of the eigenvalues less than zero and the other greater than zero one term will approach towards infinity and the other term approaches towards zero as $t \to \infty$. Overall, the point is unstable as the solutions diverge from it as time increases. This type of equilibrium point is known as *saddle point*. It is also called *hyperbolic equilibrium point*.

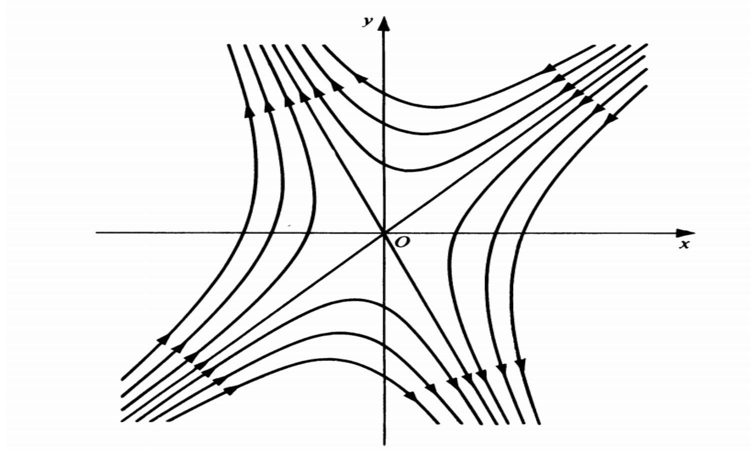

Figure 1.13: Saddle Point

Case 6: Degenerate Cases (λ_1 **or** $\lambda_2 = 0$**; both** $\lambda_1 = \lambda_2 = 0$)

In this case, the dynamics of the system is governed by higher order terms in the Taylor series expansion. Such an analysis is beyond our scope.

From the above analysis, we can appreciate that the stability of equilibrium points of nonlinear systems can be tested without actually solving the differential equation but by knowing the eigenvalues of the Jacobian matrix alone after performing suitable linearization.

1.5.5 Limit Cycles

Definition 10. (Limit Cycle). *Consider the system (1.1) when $n = 2$. A closed path C of the system (1.1) which is approached spirally from either the inside or the outside by a non-closed path C_1 of (1.1) either as $t \to +\infty$ or as $t \to -\infty$ is called a limit cycle of (1.1).*

Example 3. Consider the two-dimensional dynamical system,

$$\frac{dx}{dt} = y + x(1 - x^2 - y^2), \tag{1.25a}$$
$$\frac{dy}{dt} = -x + y(1 - x^2 - y^2). \tag{1.25b}$$

By rigorous calculations, the solution of the system becomes

$$x = \frac{\cos t}{\sqrt{1 + ce^{-2t}}},$$
$$y = -\frac{\sin t}{\sqrt{1 + ce^{-2t}}}.$$

The solutions of the system define the paths of the system in the xy plane. Examining these paths for various values of c, we note the following conclusions:
1. If $c = 0$, the path of the system is the circle $x^2 + y^2 = 1$, described in the clockwise direction.
2. If $c \neq 0$, then the paths will not be closed anymore, rather they will have a spiral behaviour.
(i) If $c > 0$, the paths are spiral lying inside the circle $x^2 + y^2 = 1$. As $t \to \infty$ they approach this circle.
(ii) If $c < 0$, the paths lie outside the circle $x^2 + y^2 = 1$. These outer paths also approach this circle as $t \to \infty$.
Since the closed path $x^2+y^2 = 1$ is approached spirally from both inside and outside by non-closed paths as $t \to \infty$, we conclude that this circle is a limit cycle of the said system. The picture is shown in the follwing:

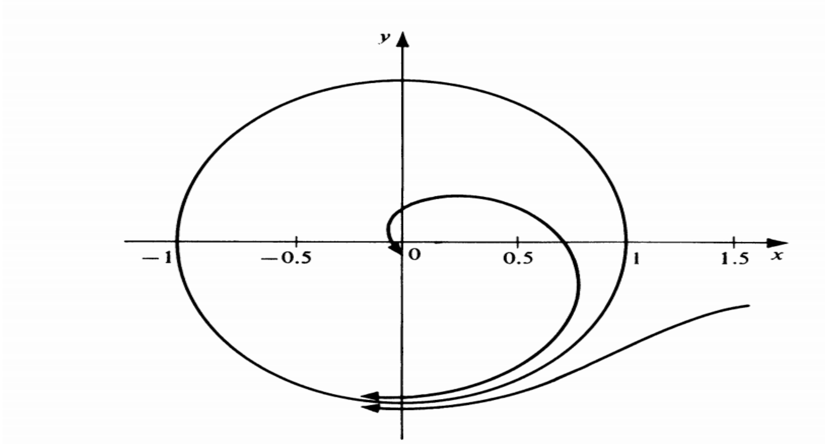

Figure 1.14: Limit Cycle

To study more about the closed orbit attractors or repellers, here about limit cycles we must assure about its existence. We state (without proof) two important theorems in the following regarding its existence:

Theorem 3. *(Poincaré-Bendixson Theorem). Consider a two-dimensional dynamical system of the form (1.20). If a solution curve $C : [x(t), y(t)]$ of the two-dimensional system remains in a domain of the (x-y) phase space in which P and Q have continuous first partial derivatives for all $t \geq \tau$, for some τ, without approaching equilibrium points, then there exists a limit cycle in the domain, and either C is a limit cycle or it approaches a limit cycle as $t \to \infty$.*

Theorem 4. *(Bendixson's Nonexistence Theorem). Consider a two-dimensional dynamical system of the form (1.20). For the system, if $s = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ does not change sign or vanish identically in a region D of the (x-y) phase space enclosed by a single closed curve, them no limit cycles lie entirely in D.*

1.5.6 Other Methods to Study Stability

Lyapunov's Method

There exists a large array of applied mathematics literature which explore dynamical systems beyond linear stability theory. Although most of this dissertation covers linear stability theory, for completeness, we introduce the Lyapunov stability criterion. First, we will start with Lyapunov function.

Definition 11. (Lyapunov Function). *Let $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with $\mathbf{x} \in X \subset \mathbb{R}^n$ be a smooth dynamical system with fixed point $\mathbf{x_0}$. Let $V : \mathbb{R}^n \to \mathbb{R}$ be a continuous function in a neighbourhood U of $\mathbf{x_0}$, then V is called a Lyapunov function for the point $\mathbf{x_0}$ if*

1. *V is differentiable in $U \setminus \{\mathbf{x_0}\}$.*
2. $V(\mathbf{x}) > V(\mathbf{x_0})$.
3. $\dot{V} \leq 0 \, \forall \, \mathbf{x} \in U \setminus \{\mathbf{x_0}\}$.

Now, in the following, we state the main theorem which connects a Lyapunov function to the stability of a fixed point of a dynamical system.

Theorem 5. *(Lyapunov Stability). Let $\mathbf{x_0}$ be a critical point of the system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, and let U be a domain containing $\mathbf{x_0}$. If there exists a Lyapunov function $V(\mathbf{x})$ for which $\dot{V} \leq 0$, then $\mathbf{x_0}$ is a stable fixed point. If there exists a Lyapunov function $V(\mathbf{x})$ for which $\dot{V} < 0$, then $\mathbf{x_0}$ is an asymptotically stable fixed point. Furthermore, if $||\mathbf{x}|| \to \infty$ and $V(\mathbf{x}) \to \infty$ for all $\mathbf{x}$, then $\mathbf{x_0}$ is said to be globally stable or globally asymptotically stable, respectively.*

So, if we can find such a *nice* function, we can easily determine the stability without any reference to a solution of the dynamical system. However, failing to find such a function does not imply instability, we simply might not have been clever enough to find one.

Centre Manifold Theory

Linear stability theory cannot determine the stability of critical points with Jacobian having eigenvalues with zero real parts. The method of centre manifold theory reduces the dimensionality of the dynamical system and the stability of this reduced system can then be investigated. The stability properties of the reduced system determine the stability of the critical points of the full system. If we start with a dynamical system in general, Centre Manifold Theory helps us to determine the stability of the trajectories associated with the eigenvalue with zero real part. In that case, we have a suitable coordinate transformation and apply relevant theorems to determine the stability of the fixed point.

1.6 Bifurcations

We have previously defined the term *bifurcation*. In this section, we will discuss different types of bifurcations namely saddle-node, pitchfork, transcritical, hopf bifurcation with various examples.

1.6.1 Saddle-Node Bifurcation

Consider the two dimensional dynamical system,

$$\dot{x} = \mu - x^2, \tag{1.26a}$$
$$\dot{y} = -y, \quad (\dot{} \equiv d/dt) \tag{1.26b}$$

where μ is a parameter. We consider different cases here.
(i) $\mu < 0$: The system has no real equilibrium points. The typical flow near the origin is illustrated in the following figure:

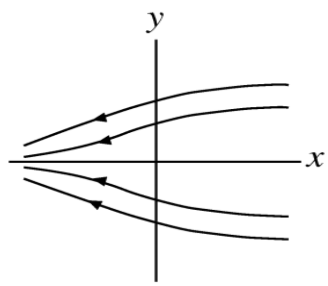

Figure 1.15: Phase trajectories of the system for $\mu < 0$

(ii) $\mu = 0$: There will be a unique equilibrium point and it will be the origin and the stability determining eigenvalues are -1, 0. So it is the non-hyperbolic case. The phase trajectories are as shown:

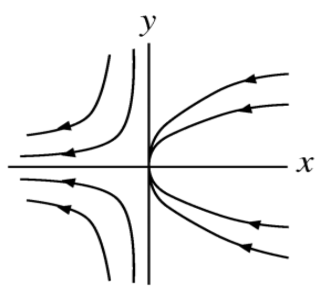

Figure 1.16: Phase trajectories of the system for $\mu = 0$

(iii) $\mu > 0$: Now we have two equilibrium points,

$$E_1\colon (x^*, y^*) = (+\sqrt{\mu}, 0),$$
$$E_2\colon (x^*, y^*) = (-\sqrt{\mu}, 0).$$

Following the previous sections, the stability determining eigenvalues for E_1 are $\lambda_1 = -1, \lambda_2 = -2\sqrt{\mu}$ and the stability determining eigenvalues for E_2 are $\lambda_1 = -1$, $\lambda_2 = 2\sqrt{\mu}$.
So from stability analysis, we can say clearly that E_1 is *stable node* and E_2 is *saddle point*. In this case, the phase trajectories are as shown:

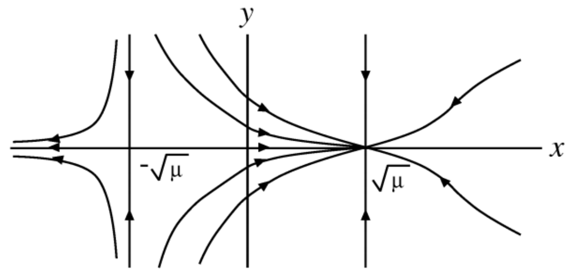

Figure 1.17: Phase trajectories of the system for $\mu > 0$

Here we can see that for a negative value of μ there is no equilibrium point but when the value of μ starts approaching 0, we have one equilibrium point and when suddenly μ surpass 0, we have two different equilibrium point. Thus for a small change of μ at 0, a bifurcation occurs giving birth to a saddle point and a node. This type of bifurcation is known as *saddle-node bifurcation.* Similarly, it can happen in arbitrary dimensions also. The associated *bifurcation diagram (x versus μ)* representing the qualitative change is as shown:

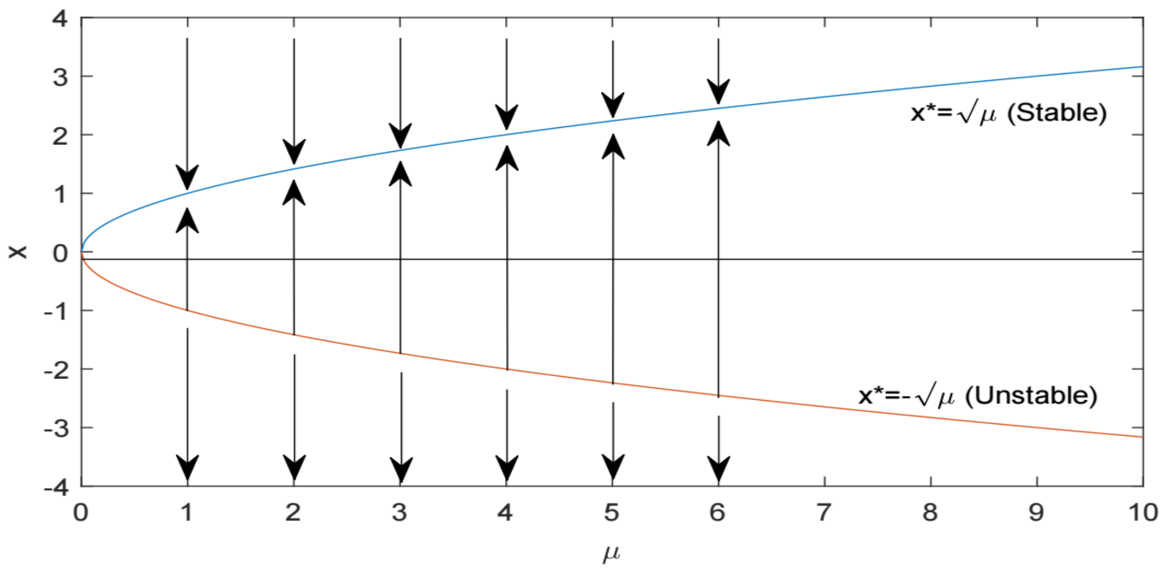

Figure 1.18: Saddle-node bifurcation diagram (arrows indicating the stable and unstable nature of the equilibrium points)

1.6.2 Pitchfork Bifurcation

Consider the dynamical system,

$$\dot{x} = \mu x - x^3, \tag{1.27a}$$
$$\dot{y} = -y, \tag{1.27b}$$

Where μ is a parameter. Now we consider different cases depending on the value of μ.
(i) $\mu < 0$:
Then (0,0) is the only equilibrium point and the eigenvalues of the Jacobian matrix corresponding to this point are $\lambda_1 = -1$, $\lambda_2 = \mu$. So it is *stable node*.
(ii) $\mu = 0$:
Then only (0,0) is one equilibrium point and the eigenvalues of the Jacobian matrix corresponding to this point are

$\lambda_1 = -1$, $\lambda_2 = 0$. So it is the non-hyperbolic case.
(iii) $\mu > 0$:
Then we have three equilibrium points namely,

$$E_0 : (x^*, y^*) = (0, 0),$$
$$E_1 : (x^*, y^*) = (-\sqrt{\mu}, 0),$$
$$E_2 : (x^*, y^*) = (\sqrt{\mu}, 0).$$

Therefore, from stability analysis, E_0 is saddle point and E_1, E_2 are stable node.
The phase trajectories are shown below:

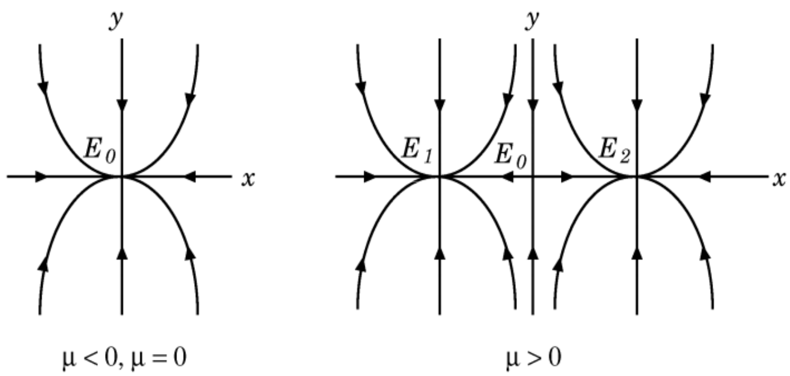

Figure 1.19: Phase trajectories corresponding to different values of μ

Therefore from the above analysis, we see as the parameter μ crosses the value 0, two new equilibrium points are born which are stable and the first equilibrium point loses its stability. Because of its tuning fork-like structure (depicted in the following diagram), it is known as *pitchfork bifurcation.*

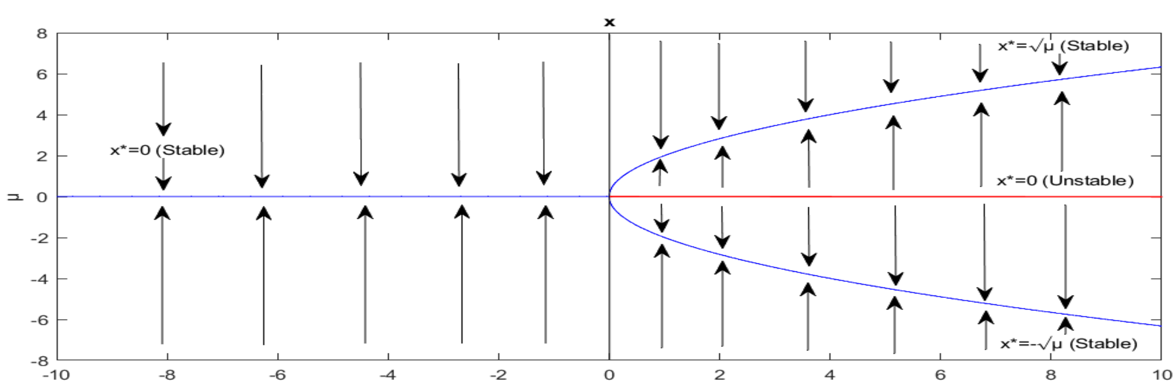

Figure 1.20: Pitchfork bifurcation diagram (arrows indicating the stable and unstable nature of equilibrium points)

1.6.3 Transcritical Bifurcation

Consider the two-dimensional dynamical system,

$$\dot{x} = -\mu x + x^2, \tag{1.28a}$$
$$\dot{y} = -y, \tag{1.28b}$$

Where μ is a parameter. Now, we consider three cases.
(i) $\mu < 0$:
In this case, we have two equilibrium points,

$$E_0 : (x^*, y^*) = (0, 0),$$
$$E_1 : (x^*, y^*) = (\mu, 0).$$

And the eigenvalues corresponding to these points are,

corresponding to E_0: $\lambda_1 = -1, \lambda_2 = -\mu$.
corresponding to E_1: $\lambda_1 = -1, \lambda_2 = \mu$.

Therefore from stability analysis, E_0 is saddle point and E_1 is stable node.
(ii) $\mu = 0$:
We have a unique equilibrium point, the origin i.e. (0,0). The eigenvalues corresponding to it are $\lambda_1 = -1, \lambda_2 = 0$. So it is the non-hyperbolic case.
(iii) $\mu > 0$:
In this case, we have two equilibrium points,

$$E_0 : (x^*, y^*) = (0, 0),$$
$$E_1 : (x^*, y^*) = (\mu, 0).$$

And the eigenvalues corresponding to these points are,

corresponding to E_0: $\lambda_1 = -1, \lambda_2 = -\mu$.
corresponding to E_1: $\lambda_1 = -1, \lambda_2 = \mu$.

Therefore from stability analysis, E_0 is stable node and E_1 is saddle point. The phase trajectories of the system corresponding to the three different conditions are given below:

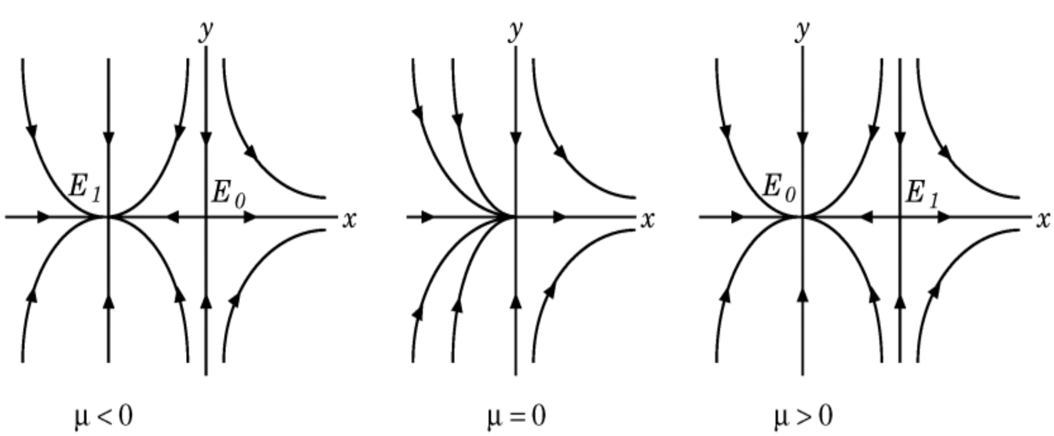

Figure 1.21: Phase trajectories corresponding to different values of μ

From the analysis we see that the stability is exchanged (or transferred) at a critical value ($\mu = 0$) when the control parameter is varied. For this reason, the bifurcation is called transcritical bifurcation or exchange of stability bifurcation and is illustrated in the following diagram.

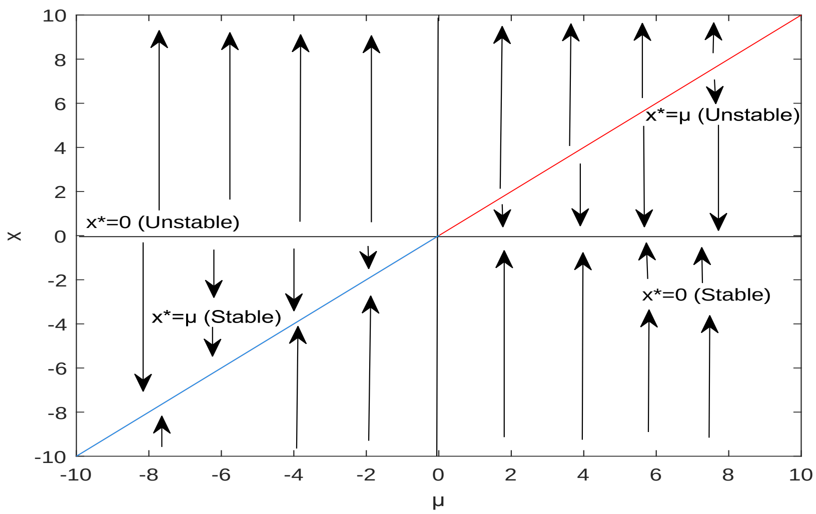

Figure 1.22: Transcritical bifurcation diagram (arrows indicating the stable and unstable nature of the equilibrium points)

1.6.4 Hopf Bifurcation

Consider the dynamical system,

$$\dot{x} = y, \tag{1.29a}$$

$$\dot{y} = b(1 - x^2)y - x, \tag{1.29b}$$

Where b is a parameter. Clearly, the equilibrium point is (0,0) and the corresponding eigenvalues of the Jacobian Matrix are $\lambda_{1,2} = \frac{b \pm \sqrt{b^2-4}}{2}$.

Now we consider different cases depending on the value of b.

(i) $-2 < b < 0$: The real part of the eigenvalues are negative. so (0,0) is a stable focus.

(ii) $b = 0$: The equilibrium point is now center and loses its stability.

(iii) $2 > b > 0$: The real part of the eigenvalues are positive and so the equilibrium point becomes the unstable focus.

Therefore from the above analysis, we see that the equilibrium point loses its stability as the parameter varies. To maintain the dynamical balance, the origin of a limit cycle happens. This type of bifurcation is known as *Hopf bifurcation.* The phase trajectories are as shown:

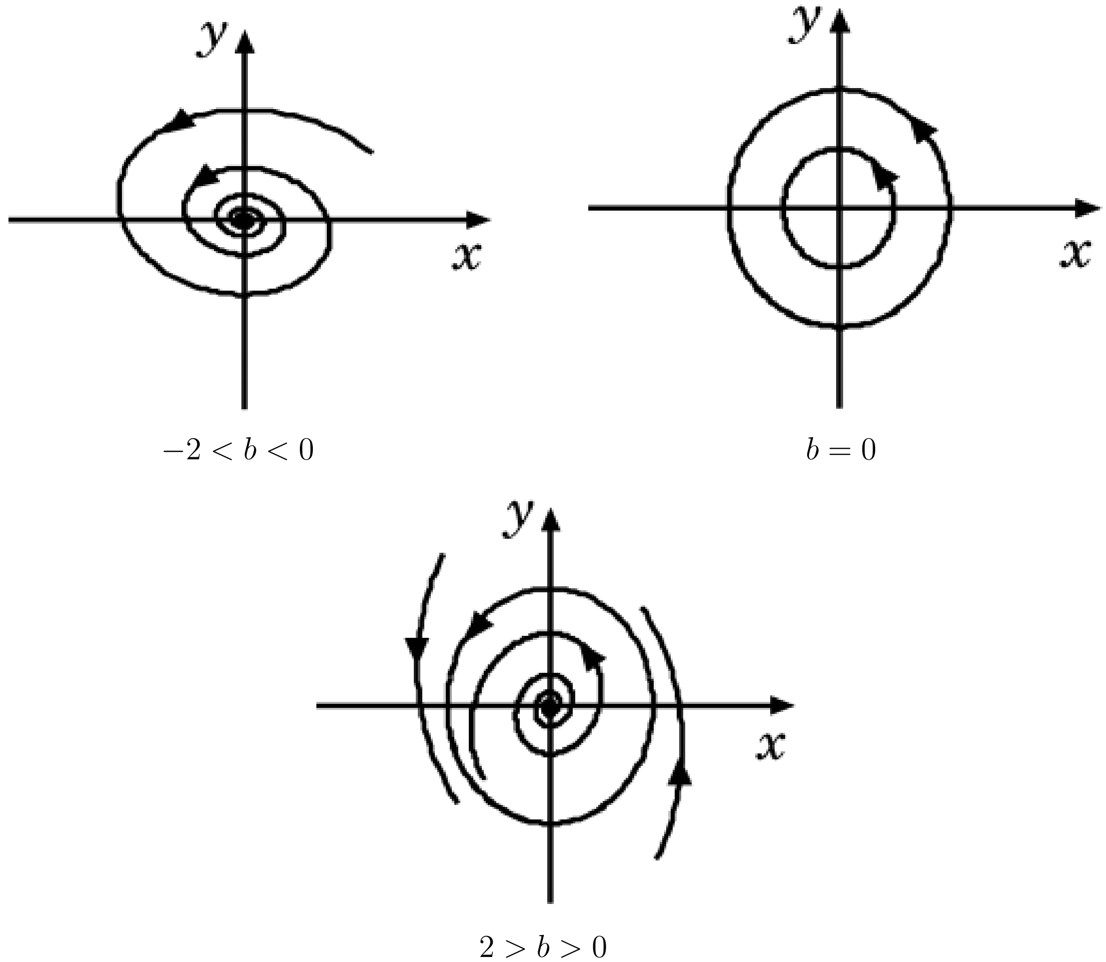

Figure 1.23: Phase trajectories for different values of b

Chapter 2

Applications of Dynamical Systems

2.1 Introduction

The goal of this chapter is to provide an overview of the applications of the powerful approach of dynamic systems to the plethora of models that have been proposed to describe the observed cosmological evolution of our universe. The mathematical theory of dynamical systems is extremely useful for understanding the global dynamics of any cosmological model, especially its late time asymptotic behaviour. Through the careful choice of the dynamical variables, a given cosmological model can be written as an autonomous system of differential equations. In this way, the analysis of the features of the phase space of this system, i.e. the analysis of the fixed points and the determination of the general behaviour of the orbits, provides insight into the global behaviour of the cosmological model. This kind of analysis is particularly useful when we are trying to establish whether a given model which presents complicated governing equations can reproduce the observed expansion of the universe. The (semi-qualitative) understanding of the dynamics of such things is the main reason behind the rapid growth of dynamical systems techniques in the last few years. In this chapter, we will be investigating the applications of dynamical systems to such models and determine their behaviours.

2.2 Application 1

2.2.1 Title of the Paper, Author(s) and Arxiv Number

- **Title of the Paper:** A Dynamical System Analysis of Three Fluid Cosmological Model.
- **Author(s):** Nilanjana Mahata and Subenoy Chakraborty. (Department of Mathematics, Jadavpur University, Kolkata-700032, West Bengal, India)
- AIP Conf. Proc., 2293, 290002 (2020)
- **Arxiv Number:** arXiv:1512.07017

2.2.2 Basic Equations

Dark matter is considered in the form of dust having energy density ρ_m and dark energy is driven by a scalar field ϕ with potential $V(\phi)$ whose energy density and pressure is given respectively by the following equations

$$\rho_d = \frac{1}{2}\epsilon\dot{\phi}^2 + V(\phi), \tag{2.1}$$

$$p_d = \frac{1}{2}\epsilon\dot{\phi}^2 - V(\phi). \tag{2.2}$$

We will have quintessence (real ϕ) if $\epsilon = 1$ or phantom field (imaginary ϕ) if $\epsilon = -1$. Here we have chosen $\epsilon = 1$. The Baryonic Matter is described by a perfect fluid with linear equation of state

$$p_b = (\nu - 1)\rho_b, \tag{2.3}$$

where ρ_b, p_b are density and pressure of the fluid and ν is the adiabatic index with $\frac{2}{3} < \nu \leq 2$.

Einstein Field Equations

The Einstein field equations can be written as (here $k = 8\pi G$ is gravitational constant, $c = 1$)

$$3H^2 = k(\rho_m + \rho_d + \rho_b), \tag{2.4}$$

$$2\dot{H} = -\frac{k}{2}(\rho_m + \rho_b + p_b + \dot{\phi^2}), \tag{2.5}$$

where H is the Hubble parameter.

Klein-Gordan Equation

Klein-Gordan equation for the scalar field is

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0. \tag{2.6}$$

Energy Conservation Equations

The energy conservation relations take the form

$$\dot{\rho_m} + 3H\rho_m = 0, \tag{2.7}$$

$$\dot{\rho_d} + 3H(\rho_d + p_d) = 0, \tag{2.8}$$

$$\dot{\rho_b} + 3H(\rho_b + p_b) = 0. \tag{2.9}$$

System of Equations

After some rigorous calculations and from the above equations, we obtain the system of equations as

$$3H^2 = k(\frac{A}{a^3} + \frac{1}{2}\dot{\phi^2} + V(\phi) + Ca^{-3}), \tag{2.10}$$

$$2\dot{H} = -\frac{k}{2}(Aa^{-3} + \nu Ca^{-3\nu} + \dot{\phi^2}), \tag{2.11}$$

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0, \tag{2.12}$$

where C and A are integration constants.

2.2.3 Formation of Autonomous System

Dimensionless Variables

To study various properties of the system, We employ the dimensionless variables

$$x = \sqrt{\frac{k}{6}}\frac{\dot{\phi}}{H}, \tag{2.13}$$

$$y = \sqrt{\frac{k}{3}}\frac{\sqrt{V(\phi)}}{H}. \tag{2.14}$$

Fractional Energy Densities

It is conventional to write fractional energy densities as

$$\Omega_m = \frac{kAa^{-3}}{3H^2}, \tag{2.15}$$

$$\Omega_b = \frac{kCa^{-3\nu}}{3H^2}. \tag{2.16}$$

Autonomous System

The evolution of the dynamical system is described by the autonomous system

$$\frac{dx}{dN} = 3x(x^2 - 1 + \frac{\Omega_m}{2} + \frac{\nu\Omega_b}{2}) - \sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)}y^2, \tag{2.17a}$$

$$\frac{dy}{dN} = y[3(x^2 + \frac{\nu\Omega_b}{2} + \frac{\Omega_m}{2}) + \sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)}x], \tag{2.17b}$$

$$\frac{d\Omega_m}{dN} = -3\Omega_m[1 - \Omega_m - 2x^2 - \nu\Omega_b], \tag{2.17c}$$

where $N \equiv ln\ a$.

Other Parameters

The effective equation of state for this three fluid model has the expression,

$$\omega_{eff} = \frac{p_d + p_b}{\rho_m + \rho_d + \rho_b} = -1 + 2x^2 + \nu(1 - \Omega_m - x^2 - y^2) + \Omega_m. \tag{2.18}$$

The equation of state of dark energy is given by

$$\omega_\phi = \frac{p_d}{\rho_d} = \frac{x^2 - y^2}{x^2 + y^2}, \tag{2.19}$$

and the density parameter for the scalar field is

$$\Omega_\phi = \frac{k\rho_\phi}{3H^2} = x^2 + y^2. \tag{2.20}$$

2.2.4 Phase Space Analysis and Critical Points

A. 3D Autonomous System

We assume $\frac{V'(\phi)}{V(\phi)}$ = constant. So, we can write $\sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)}$ = constant = α (say). Now, considering the linear perturbations about the critical point $(x_c,\ y_c,\ \Omega_{mc})$ into the dynamical system (2.17) the linearized perturbation matrix takes the form

$$\begin{bmatrix} \begin{matrix} 9x_c^2(1-\frac{\nu}{2}) - 3(1-\frac{\Omega_{mc}}{2}) \\ +3(1-\Omega_{mc})\frac{\nu}{2} \end{matrix} & -3\nu y_c x_c - 2\alpha y_c & \frac{3x_c}{2}(1-\nu) \\ 3y_c x_c(2-\nu) + \alpha y_c & \begin{matrix} 3x_c^2(1-\frac{\nu}{2}) + \frac{3\nu}{2} + \frac{3\Omega_{mc}}{2}(1-\nu) \\ +\alpha x_c - \frac{9}{2}\nu y_c^2 \end{matrix} & \frac{3x_c}{2}(1-\nu) \\ -6x_c\Omega_{mc}(\nu-2) & -6\nu y_c\Omega_{mc} & \begin{matrix} 6\Omega_{mc}(1-\nu) + 3x_c^2(2-\nu) \\ -3\nu y_c^2 - 3(1-\nu) \end{matrix} \end{bmatrix}$$

The hyperbolic critical points of the system, related cosmological parameters, eigen values are given in the following TABLE I.

TABLE I. *Equilibrium Points, related parameters and Eigen values*

Equilibrium Point	x	y	Ω_m	Ω_ϕ	Ω_b	ω_{eff}	Eigen values
C_1	0	0	0	0	1	$-1+\nu$	$-3(1-\nu), \frac{3\nu}{2}, 3(\frac{\nu}{2}-1)$
C_2	1	0	0	1	0	1	$6(1-\frac{\nu}{2}), 3+\alpha, 3$
C_3	-1	0	0	1	0	1	$6(1-\frac{\nu}{2}), 3-\alpha, 3$
C_4	0	0	1	0	0	0	$-\frac{3}{2}, \frac{3}{2}, 3(1-\nu)$
C_5	$\frac{3}{2\alpha}(\nu-1)$	$\pm\frac{3}{2\alpha}(1-\nu)$	$< 1-\frac{9}{2}\frac{(1-\nu)^2}{\alpha^2}$	$\frac{9}{2}\frac{(1-\nu)^2}{\alpha^2}$	>0	0	in TABLE II

For different values of ν, α, C_5 will denote different critical points. These points are listed in TABLE II with their Eigen Values.

TABLE II. *C_5 for different ν and α, related parameters and Eigen values*

Equilibrium Point	α	ν	x	y	Ω_m	Ω_ϕ	Ω_b	ω_{eff}	Eigen values
C_{5a}	-2.028	1.8	-0.5916	0.5916	0.3	0.699	0.001	0	-0.0096 $\pm$ 2.0543i,-0.2356
C_{5b}	-1.52	1.6	-0.5916	0.5916	0.3	0.699	0.001	0	-0.027 $\pm$ 1.8655i, -0.0055
C_{5c}	2	1.3	0.225	-0.225	0.898	0.101	0.001	0	-0.9939 $\pm$ 0.4031i, 1.6365
C_{5d}	1.3	1.2	0.375	-0.375	0.718	0.281	0.001	0	-0.7040 $\pm$ 0.7392i, 1.2323
C_{5e}	-0.5	0.8	0.57	0.57	0.35	0.649	0.001	0	-0.527 $\pm$ 0.594i, -0.3735
C_{5f}	-1.5	0.9	0.1	-0.1	0.98	0.02	0	0	-1.446, 1.304, -0.3055
C_{5g}	-0.5	1.2	-0.6	0.6	0.28	0.72	0	0	-0.1174 $\pm$ 1.4189i, 0.1121
C_{5h}	1.5	0.75	-0.25	0.25	0.87	0.125	0.005	0	-0.828 $\pm$ 0.18i, -1.21

B. 2D Autonomous System

The 3D System (2.17) can be reduced to a 2D Autonomous System if we do not consider the time evolution of dark matter. We consider Ω_m to be a constant lying within $0.24 < \Omega_m < 0.34$. Then the system becomes

$$\frac{dx}{dN} = 3x[x^2 - 1 + \frac{\Omega_m}{2} + \frac{\nu}{2}(1-\Omega_m - x^2 - y^2)] - \alpha y^2, \tag{2.21a}$$

$$\frac{dy}{dN} = y[3x^2 + 3\frac{\nu}{2}(1-\Omega_m - x^2 - y^2) + 3\frac{\Omega_m}{2}) + \alpha x]. \tag{2.21b}$$

Different critical points are obtained for different values of α, ν. These points with there nature and related parameters are listed in the tables III to IX and the corresponding phase portraits are given in the following.

TABLE III. *Critical points for $\Omega m = 0.32$, $\nu = 1.9$, $\alpha = 1.7$, related parameters*

Equilibrium Point	x	y	Ω_ϕ	ω_{eff}	Nature
C_{2a}	0	0	0	0.61	Saddle
C_{2b}	-0.44	0.76	0.77	-0.47	Stable Node
C_{2c}	-0.44	-0.76	0.77	-0.47	Stable Node
C_{2d}	1.58	0	2.50	0.86	Unstable Node
C_{2e}	-1.48	0	2.19	0.83	Unstable

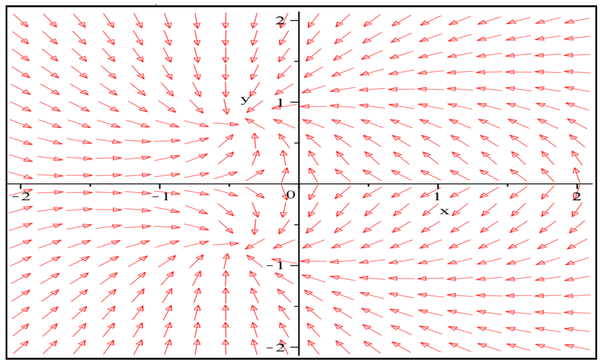

Figure 2.1: Phase portrait for the system, related table III

TABLE IV. *Critical points for* $\Omega m = 0.3$, $\nu = 1.8$, $\alpha = -2$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	ω_{eff}	Nature
C_{2a}	0	0	0	0.56	Saddle
C_{3b}	0.52	0.70	0.76	-0.27	Stable Node
C_{3c}	0.52	-0.70	0.76	-0.27	Stable Node

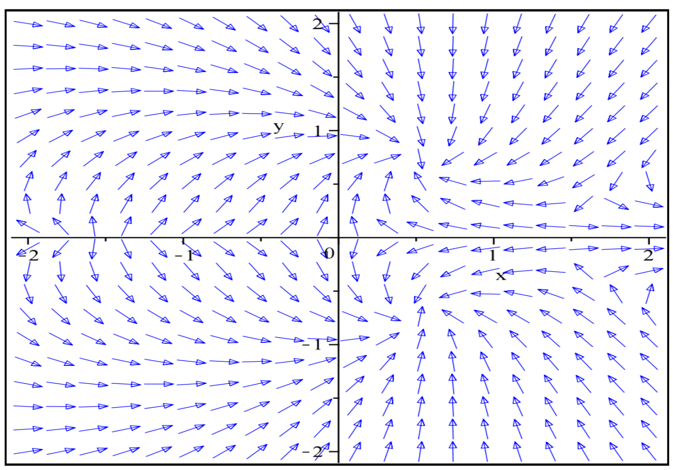

Figure 2.2: Phase portrait for the system, related table IV

TABLE V. *Critical points for* $\Omega m = 0.2$, $\nu = 1.6$, $\alpha = 1.1$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	Ω_b	ω_{eff}	Nature
C_{2a}	0	0	0	0.8	0.48	Saddle
C_{4b}	-0.33	0.89	0.90	0.059	-0.74	Stable Node
C_{4c}	-0.33	-0.89	0.90	0.059	-0.74	Stable Node

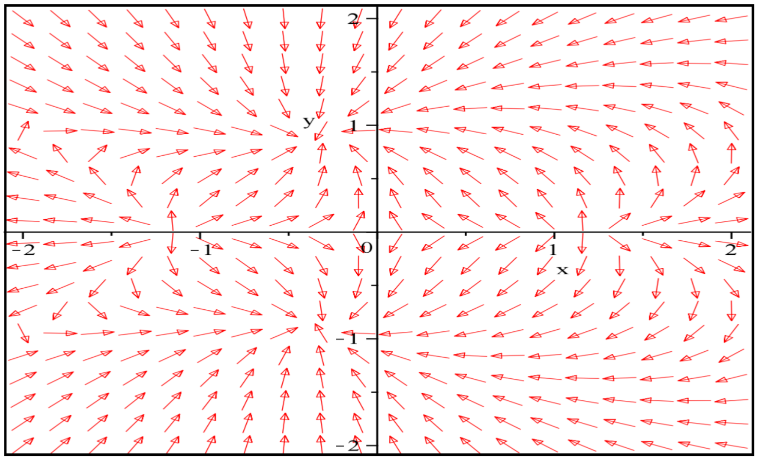

Figure 2.3: Phase portrait for the system, related table V

TABLE VI. *Critical points for* $\Omega m = 0.22$, $\nu = 1.2$, $\alpha = 1.1$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	Ω_b	ω_{eff}	Nature
C_{2a}	0	0	0	0.78	0.156	Saddle
C_{5b}	-0.32	0.84	0.81	0.001	-0.61	Stable Node
C_{5c}	-0.33	-0.89	0.90	0.001	-0.61	Stable Node

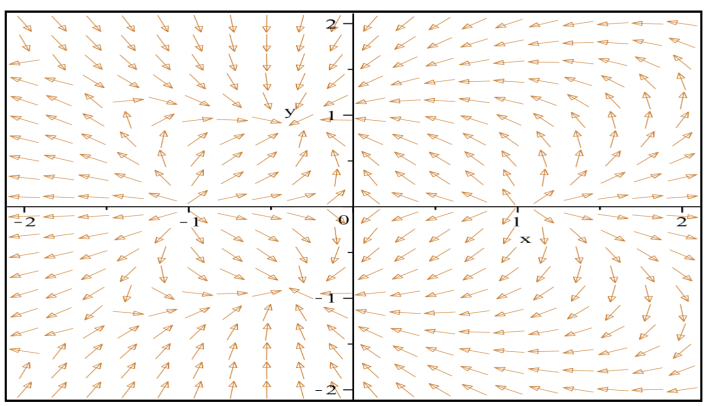

Figure 2.4: Phase portrait for the system, related table VI

TABLE VII. *Critical points for* $\Omega m = 0.24$, $\nu = 1.3$, $\alpha = 3$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	Ω_b	ω_{eff}	Nature
C_{2a}	0	0	0	0.76	0.228	Saddle
C_{6b}	-0.58	0.47	0.56	0.20	0.18	Stable Node
C_{6c}	-0.58	-0.47	0.56	0.20	0.18	Stable Node

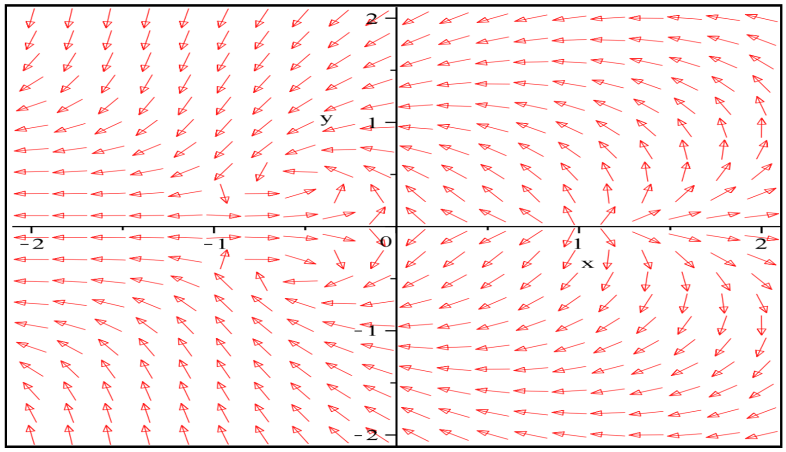

Figure 2.5: Phase portrait for the system, related table VII

TABLE VIII. *Critical points for* $\Omega m = 0.24$, $\nu = 1.3$, $\alpha = 1.3$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	Ω_b	ω_{eff}	Nature
C_{2a}	0	0	0	0.76	0.228	Saddle
C_{7b}	-0.40	0.87	0.92	0.001	-0.64	Stable Node
C_{7c}	-0.40	-0.87	0.92	0.001	-0.64	Stable Node

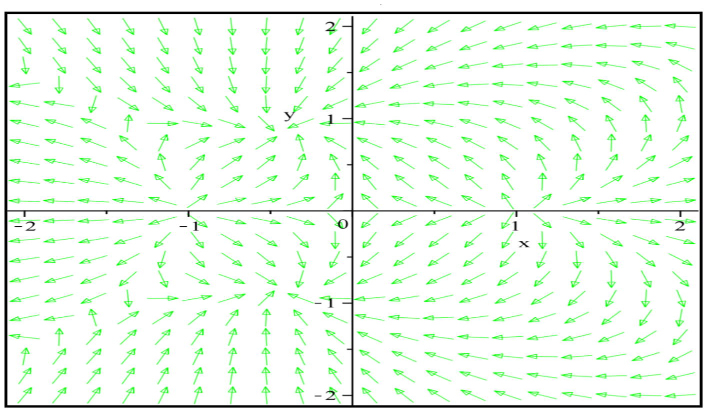

Figure 2.6: Phase portrait for the system, related table VIII

TABLE IX. *Critical points for* $\Omega m = 0.2$, $\nu = 1.6$, $\alpha = -1.1$, *related parameters*

Equilibrium Point	x	y	Ω_ϕ	Ω_b	ω_{eff}	Nature
C_{2a}	0	0	0	0.8	0.48	Saddle
C_{4b}	0.33	0.89	0.90	0.001	-0.74	Stable Node
C_{4c}	0.33	-0.89	0.90	0.001	-0.74	Stable Node

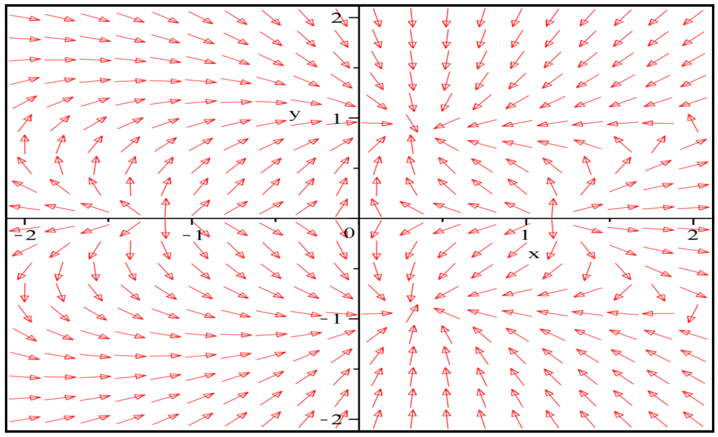

Figure 2.7: Phase portrait for the system, related table IX

2.3 Application 2

2.3.1 Title of the Paper, Author(s) and Arxiv Number

- **Title of the Paper:** Dynamical System Analysis of Brane Induced Gravity with Tachyon Field.
- **Author(s):** A. Ravanpak and G.F. Fadakar. (Department of Physics, Vali-e-Asr University, Rafsanjan, Iran)
- Class. Quantum Grav., 36, 235003 (2019)
  DOI 10.1088/1361-6382/ab4eec
- **Arxiv Number:** arXiv:1906.04465

2.3.2 The Model

The Friedmann equation on the brane in the normal branch of DGP scenario is

$$H^2 + \frac{H}{r_c} = \frac{1}{3M_p^2}(\rho_m + \rho_{tac}). \tag{2.22}$$

Here, r_c is crossover scale (relation between the 4D and 5D Planck mass), H is the Hubble parameter, ρ_m is the matter energy density, ρ_{tac} denotes the density of the tachyon field, M_p is the 4D Planck mass. The energy density and pressure of tachyon field are expressed as

$$\rho_{tac} = \frac{V(\phi)}{\sqrt{1-\dot{\phi}^2}}, \tag{2.23}$$

$$P_{tac} = -V(\phi)\sqrt{1-\dot{\phi}^2}, \tag{2.24}$$

in which ϕ and $V(\phi)$ are tachyon field trapped on the brane and the tachyon potential respectively and dot means derivative with respect to the cosmic time. The conservation equations are

$$\dot{\rho} + 3H\rho_m = 0, \tag{2.25}$$

$$\dot{\rho}_{tac} + 3H(\rho_{tac} + P_{tac}) = 0. \tag{2.26}$$

Now, from the above equations, one can obtain the equation of motion of the tachyon field as

$$\frac{\ddot{\phi}}{1-\dot{\phi}^2} + 3H\dot{\phi} + \frac{V_\phi}{V} = 0, \tag{2.27}$$

in which V_ϕ represents the derivative of the $V(\phi)$, with respect to the tachyon scalar field.

2.3.3 Phase Space and Stability Analysis

We define a set of dimensionless dynamical variables

$$x^2 = \frac{\rho_m}{3M_p^2(H^2 + \frac{H}{r_c})}, \tag{2.28}$$

$$y^2 = \frac{V}{3M_p^2(H^2 + \frac{H}{r_c})}, \tag{2.29}$$

$$d = \dot{\phi}, \tag{2.30}$$

$$z^2 = 1 + \frac{1}{Hr_c}. \tag{2.31}$$

Using the newly introduced variables, the Friedmann constraint (2.22) reads

$$x^2 + \frac{y^2}{\sqrt{1-d^2}} = 1, \tag{2.32}$$

and the Raychaudhury equation can be recast as

$$\frac{\dot{H}}{H} = \frac{-3z^2}{z^2+1}(x^2 + y^2\frac{d^2}{\sqrt{1-d^2}}). \tag{2.33}$$

The set of evolution equations can be obtained as

$$d' = -(3d - \sqrt{3}zy\lambda)(1-d^2), \tag{2.34a}$$

$$y' = -\frac{\sqrt{3}}{2}y^2dz\lambda + \frac{3}{2}y(1 - y^2\sqrt{1-d^2}), \tag{2.34b}$$

$$z' = \frac{3}{2}\frac{z(z^2-1)}{z^2+1}(1 - y^2\sqrt{1-d^2}). \tag{2.34c}$$

Here, prime means derivative with respect to ln a, and also $\lambda = -M_pV_\phi/V^{3/2}$. It is convenient to divide our discussion into two parts depending on the parameter λ. We consider two general cases:

1. $\lambda =$ constant.

2. $\lambda = \lambda(\phi)$. Various potentials lead to a varying λ.

The case $\lambda =$ constant

TABLE X. *The fixed points of the model for* $\lambda =$ constant

Point	(d, y, z)	Eigenvalues	w_{tot}	Stability
P_1	(0, 0, 1)	(-3, 3/2, 3/2)	0	Saddle
$P_2^{\pm}$	($\pm$0.84, $\pm$0.73, 1)	(-3.7, -1.9, -1.9)	-0.28	Stable
$P_3^{\pm}$	($\pm$1, 0, 1)	(6, 3/2, 3/2)	0	Unstable

The case $\lambda = \lambda(\phi)$

In this case we can first find the critical points of the system and then discuss their properties. The results have been given in TABLE XI, in which $y^* = \sqrt{\frac{\alpha}{6}}$, $d^* = \lambda y^*/\sqrt{3}$, $\alpha = \sqrt{\lambda^4 + 36} - \lambda^2$ and $\delta = -3 + \frac{\lambda^2}{12}y^{*2}$.

TABLE XI. *The fixed points of the model for $\lambda = \lambda(\phi)$*

Point	(d, y, z)	Eigenvalues	w_{tot}	Stability	$\lambda(\phi)$
P_1	(0, 0, 1)	(-3, 3/2, 3/2)	0	Saddle	any
$P_2^{\pm}$	$(\pm d^*, \pm y^*, 1)$	(-6(3 + δ), δ, δ)	$-\frac{\alpha}{6}\sqrt{1-\frac{\alpha\lambda^2}{18}}$	Stable	any
$P_3^{\pm}$	($\pm$1, 0, 1)	(6, 3/2, 3/2)	0	Unstable	any
$P_4^{\pm}$	(0, $\pm$1, z)	(0, -3, -3)	-1	Stable	0

In the following, we have depicted several 2D phase trajectories for different values of λ and treating the other variable constant.

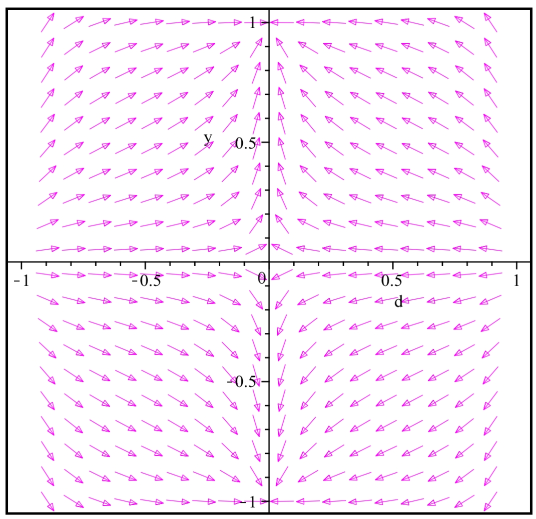

Figure 2.8: The 2D phase space (y, d) for $\lambda = 0$, treating z as constant. (of any value, in this case)

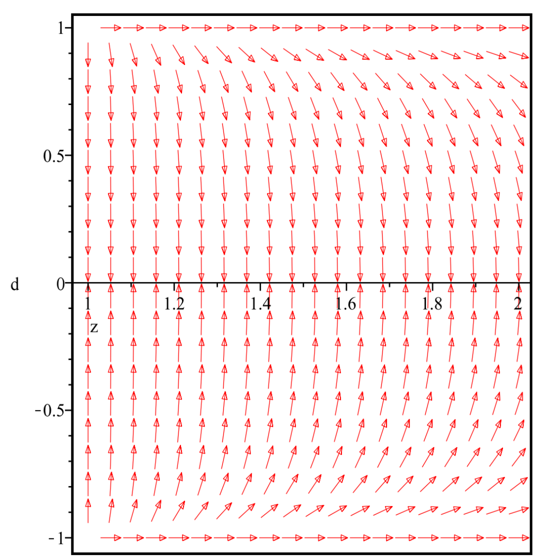

Figure 2.9: The 2D phase space (d, z), for $\lambda = 0$ and $y = \pm 1$.

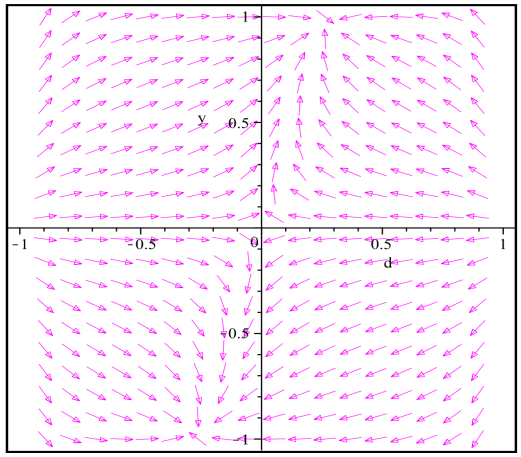

Figure 2.10: The 2D phase space (y, d), for $\lambda = 0.1$, $z = 5$.

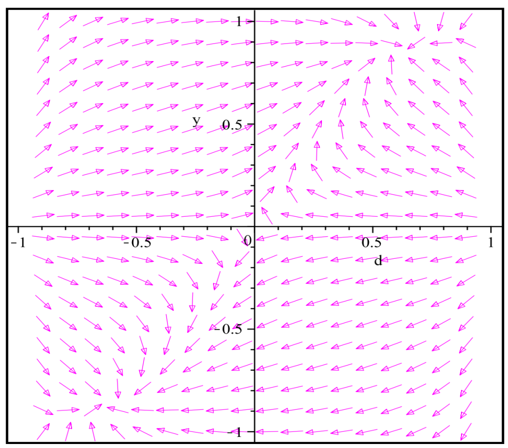

Figure 2.11: The 2D phase space (y, d) for $\lambda = 1.4$, $z = 0.9$.

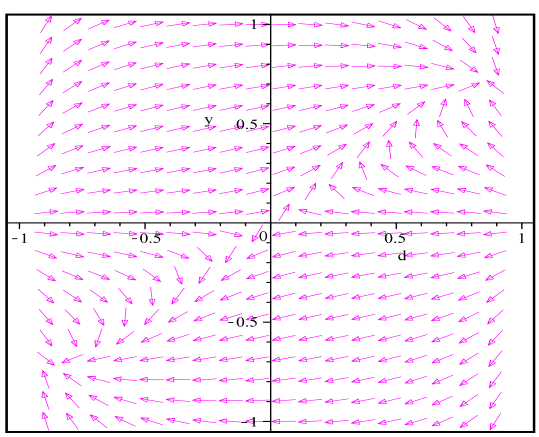

Figure 2.12: The 2D phase space (y, d) for $\lambda = 2.4$, $z = 0.9$.

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